Audio file

Uncertainty - Definition and Types.m4a

Transcript

Hi there, my name is Amelis Beaubelin and welcome to this video in the series Entrepreneurship in Action Ideation. This is the first video in the series into action dealing with uncertainty and the objective of this video is to introduce you to a definition of uncertainty. We talk about how the definition of uncertainty and how uncertainty is different from risk. And next we move on with discussing different types of uncertainty. Now, before we move to a definition of what uncertainty is about, it's important to emphasize why uncertainty is such a central concept in entrepreneurship. As an entrepreneur, especially as a high-tech entrepreneur, you will soon realize that uncertainty permeates every decision of your entrepreneurial life. When you're deciding to bring a novel technology to market, it's hard to predict whether that technology will be able to deliver the functionalities you have in mind. It's hard to predict whether you will be able to build a product that is attractive to a certain customer segment. It's uncertain whether you will secure sufficient funding. In all, there are a lot of uncertainties. And as an entrepreneur, it will be crucial to develop the capabilities to deal with uncertainty. Let's now move to a definition. What is uncertainty really about? The best way to explain uncertainty is by drawing an analogy with a lottery. Let's have a look at the first urn, a predictive urn. If you have to draw a ball from this urn, it's predictable what your chances are of success. Say success is drawing a green ball. If you look at the urn, you can see that in total there are 10 balls and only four green balls in it. So that means you have a 40% chance of being successful. That means the distribution of potential events to happen in the future is known, but you do not know your particular drop. That means prediction is possible. If we then have a look at the risky urn, we see that the situation is already pretty different. There we are confronted with an opaque urn. We do not know upfront what is inside, so the distribution is unknown. However, if we start drawing balls from the urn, we see that the first draw is a red ball. Then we move on to another red ball. Then we have a green ball, a red ball and again a green ball. So after a while, we come to understand what is inside of the urn. And we see that apparently there are only two options. Either you draw a green ball or either you draw a red ball. And experience also teaches us that there are apparently more red balls in the urn than green balls. So that means that although it's difficult to predict what will happen in the future, experience helps. And after a while, it becomes more predictable what you can expect. Then a truly uncertain situation or the uncertain urn is a following one. There we still have an opaque urn, so we do not know upfront what is inside. We cannot see it. And the moment we start drawing something from the

urn, every time we're confronted with something different, as is illustrated with the icons here on the slide. So that means that the future is unknown up front. We do not know what can happen. There are so many different things that could be happening, and it's impossible to know up front what it would be. And what is even more problematic compared to the risky urn, experience does not help. So the moment we start throwing balls or elements from this urn, it does not help us to predict what might be coming next. So that's a situation of true uncertainty, also referred to as 90 uncertainty. Now we have an understanding of what uncertainty really is about and how it is different from risk. Let's have a look at the different sources of uncertainty. There are many classifications of uncertainty, but here we stick to the one introduced by Kapoor and Klutter because this is very relevant to high-tech entrepreneurs. If you're bringing a novel technology to market, of course, the first source of uncertainty is your technology. As an entrepreneur, you do not know upfront whether you will be able to deliver a fully functional novel technology. And it's also hard to predict how many resources you need, how much time it will take you to deliver a new product building on novel technology. So there's a lot of uncertainty about the technology itself. Another source of uncertainty relates to application uncertainty. Application uncertainty is that, as an entrepreneur, you do not know upfront what will be the best application for that particular technology. Let's consider, for example, drones. Everyone understands the main functionalities of a drone, but if you want to earn money with your drone, there are many purposes for which you can use that drone. You can use the drone in medical urgencies, you can use the drone for delivering packages, you can just use it for hobbyists. So there are many different situations in which you could use the drone, but up front it's hard to predict which one will be the most successful one. And that brings us to a third source of uncertainty, user uncertainty. So even if you think, let's go for the package delivery option with our drone, it's hard to predict up front what potential customers or users would expect from your drone. What is like the maximum weight your drone should be able to carry? There are a lot of uncertainties about the benefits and the expectations the users might have about your technology. Then a fourth type of uncertainty is ecosystem uncertainty. Ecosystem uncertainty refers to the fact that at one point in time if you deliver an integrated solution to the market, you probably have to set up an ecosystem. And it's an ecosystem where different partners, suppliers collaborate with you to build that integrated solution. Perhaps you need to collaborate with a software developer that develops the software for mapping the root of the drone. So upfront, it's hard to predict who you will collaborate with and what the terms of that collaboration will be like. And that brings us then to our final source of uncertainty, business model uncertainty. Business model uncertainty is about the profit equation. It's about being unable to predict upfront whether you will be earning money with your innovation. Even if you have an idea on how to build a product, even if you have an idea of the type of customers that would be interested in it, It's upfront, difficult to predict, even impossible, whether you will be able to do so in a way that allows you to capture some

value from it and earn back your investment in the end. So that's business model uncertainty. To conclude, what have we learned with this video? First of all, it's important to realize that uncertainty permeates every decision of the professional life of an entrepreneur and as an entrepreneur, you need to be able to deal with it. And tech entrepreneurs do not only deal with technological uncertainty, they deal with a variety of uncertainty sources, and which we have seen was user uncertainty, application uncertainty, ecosystem uncertainty, and business model uncertainty. So many sources of uncertainty that affect the life of an entrepreneur. That's it for this video. Thanks for watching.